Deciding What to Try Next: An Action Plan 🗺️

When your machine learning model has unacceptably large errors, the bias/variance diagnosis provides a clear guide on what to do. The six most common actions to take each map to fixing either high bias or high variance.

Fixing a High Variance Problem (Overfitting)

If your diagnosis is **high variance** ($J_{cv} \gg J_{train}$), your model is too complex and is failing to generalize. Your goal is to **simplify the model** or provide it with more data to learn the general pattern.

- **1. Get more training examples:**
  - As seen in the learning curves, adding more data is one of the most effective ways to fix high variance. It makes it harder for a complex model to overfit the noise and forces it to learn the true underlying pattern.
- **2. Try a smaller set of features:**
  - Having too many features gives the model too much flexibility to overfit.
  - By reducing the number of features (feature selection), you reduce the model's complexity and its ability to fit a "wiggly" function to the training data.
- **3. Increase the regularization parameter (λ):**
  - A larger λ increases the penalty on the parameters, forcing them to be smaller.
  - This results in a simpler, smoother function that is less prone to overfitting.

Fixing a High Bias Problem (Underfitting)

If your diagnosis is **high bias** (J_{train} is high), your model is too simple to even capture the patterns in the training data. Your goal is to **make the model more complex** or give it more information to work with.

- **1. Get additional (new) features:**
  - If your model is trying to predict a house price *only* from its size, it will have high bias. It's missing crucial information.
  - Adding new, relevant features (like the number of bedrooms, age of the house, etc.) gives the model more information to make a good prediction.
- **2. Add polynomial features (e.g., x^2, x^3, $x \cdot y$):**
  - This is a form of feature engineering that increases model complexity.
  - It allows the model to fit a non-linear curve instead of just a straight line, which can significantly reduce bias.

- **3. Decrease the regularization parameter (λ):**
    - A smaller λ *reduces* the penalty on the parameters.
    - This gives the model more flexibility to fit a more complex, "wiggly" curve to the training data, thereby reducing bias.

> ***Note:*** *You should **not** try to fix high bias by* reducing *your training set size. While this would lower J_{train} (as seen in the learning curves), it would make your model's generalization (CV) error worse.*

Summary of Actions

Diagnosis	Your Goal	What to Try Next
High Variance (Overfitting)	Simplify the model or get more data.	1. **Get more training examples.** 2. **Try a smaller set of features.** 3. **Increase λ.**
High Bias (Underfitting)	Make the model more complex or get more features.	1. **Get additional features.** 2. **Add polynomial features.** 3. **Decrease λ.**